

Multivariate Analysis of the Doubs River Fish-Environment System

Generated from the BCB743 Quantitative Ecology data

07 July 2026

Abstract

The Doubs River fish and environmental data were analysed with the main multivariate methods of the Quantitative Ecology chapters, namely PCA, CA, DCA, PCoA, nMDS, RDA, CCA, and db-RDA, and then extended with variable-selection, spatial, beta-diversity, and species-response analyses. The methods agree on one dominant structure, a longitudinal river-continuum gradient from cool, steep, oxygen-rich headwaters to lower, slower, nutrient- and organic-load-enriched reaches, and fish composition follows it as an ordered replacement of trout-zone species by lowland cyprinids. Two refinements emerge from the extended analyses. A short mid-river reach departs from the continuum as an organic-pollution oxygen sag that collapses local richness and keeps oxygen and biological oxygen demand in the selected models, and the contrast between data-driven and theory-driven variable selection shows that the retained oxygen variables are proximate correlates of an upstream nutrient load rather than independent causes. Constrained models explain about half of the community variation, and beta-diversity partitioning attributes most of the turnover to species replacement.

1 Data and Pre-processing

The source files are the local BCB743 Doubs River tables. Environmental PCA used all 30 environmental rows. Species ordinations and constrained analyses removed site 8 because it has zero total fish abundance, which leaves 29 matched fish, environmental, and spatial rows.

Table 1: Local Doubs River data files used in this report.

Table	Role	Rows	Columns	Notes
DoubsEnv.csv	Environmental predictors	30	11	Used for PCA; matched subset used for direct gradient analyses
DoubsSpe.csv	Fish abundance response	30	27	Site 8 has zero total abundance and was removed for species ordinations
DoubsSpa.csv	Spatial coordinates	30	2	Used for the site map

Table 2: Environmental variables and their longitudinal behaviour. The last column is the Pearson correlation with distance from source (‘dfs’) across all 30 sites.

Code	Variable	Minimum	Median	Maximum	r with dfs
dfs	Distance from source	0.30	175.20	453.00	1.00
alt	Altitude	172.00	395.00	934.00	-0.94
slo	Channel slope	0.20	1.20	48.00	-0.38
flo	Flow rate	0.84	22.10	69.00	0.95
pH	pH	7.70	8.00	8.60	0.01
har	Water hardness	40.00	89.00	110.00	0.70
pho	Phosphate	0.01	0.29	4.22	0.48
nit	Nitrate	0.15	1.60	6.20	0.75
amm	Ammonium	0.00	0.10	1.80	0.41
oxy	Dissolved oxygen	4.10	10.20	12.40	-0.51
bod	Biological oxygen demand	1.30	4.15	16.70	0.39

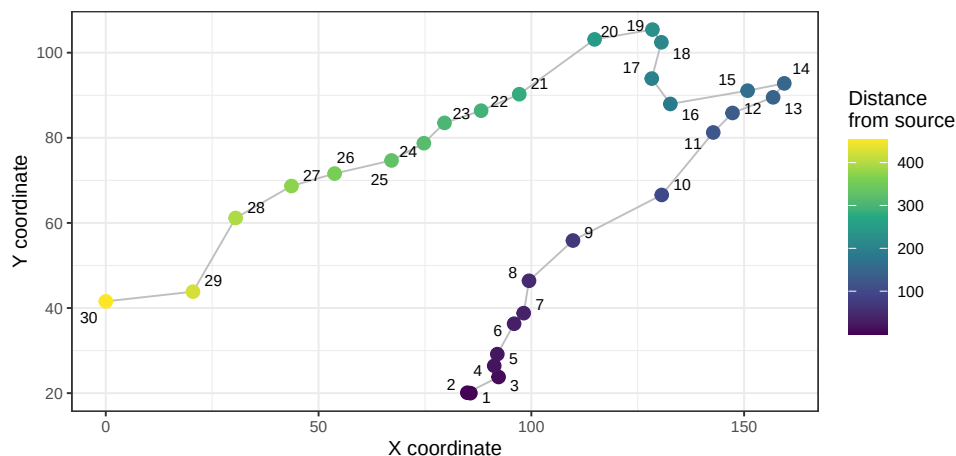


Figure 1: Spatial configuration of the Doubs River sites. Colour gives distance from source.

2 Methods

The methods follow the BCB743 ordination sequence and then extend it with a set of more advanced checks, each asking a sharper question of the same data. Unconstrained ordinations come first. They let the data reveal their dominant structure without forcing environmental predictors into the solution. PCA summarises covariance among the measured environmental variables, CA and DCA summarise unimodal fish turnover, and PCoA and nMDS summarise Bray-Curtis dissimilarities among sites. Constrained ordinations come second, and they ask a sharper question, namely how much of the fish-composition pattern the measured environmental variables can account for. RDA, CCA, and db-RDA answer that question under three different geometries. The advanced analyses come last. They ask whether the same conclusion survives when the problem is reframed in terms

of collinearity, spatial structure, distributional assumptions, the philosophy of variable selection, and the shape of individual species responses.

Table 3: Summary of methods and headline outputs.

Method	Response	Key output	Interpretation
PCA	11 environmental variables	PC1 = 54.3%, PC2 = 19.7%	Environmental covariance forms a strong river-continuum axis.
CA	27 fish abundance columns	CA1 = 51.5%, CA2 = 12.4%	Species composition turns over mainly along the river continuum.
DCA	27 fish abundance columns	DCA1 axis length = 3.86	Gradient length supports a long unimodal species gradient and explains the CA arch.
PCoA	Bray-Curtis dissimilarity of fish assemblages	MDS1 = 52.4% of positive inertia	Bray-Curtis distances recover the same faunal turnover.
nMDS	Ranks of Bray-Curtis dissimilarities	Stress = 0.075	Rank-order composition has a faithful two-dimensional representation.
RDA	Hellinger-transformed fish abundances	Selected: dfs, oxy, bod; adj. $R^2 = 0.54$	Linear constrained model confirms environment-composition coupling.
CCA	Raw fish abundance table	Selected: dfs, oxy, alt; adj. $R^2 = 0.50$	Unimodal constrained model confirms the same coupling.
db-RDA	Bray-Curtis dissimilarity of fish assemblages	Selected: dfs, bod, oxy; adj. $R^2 = 0.57$	Distance-based constrained model gives the clearest community-distance test.

3 Unconstrained Gradients

3.1 Environmental PCA

The standardised environmental PCA is strongly one-dimensional: PC1 explains 54.3% of the environmental variance and PC2 explains 19.7%. PC1 contrasts high-altitude, high-oxygen, steep upper sites against downstream sites with higher distance from source, discharge, nitrate, phosphate, ammonium, hardness, and biological oxygen demand.

Table 4: Environmental PCA eigenvalues.

Axis	Eigenvalue	Percent	Cumulative
PC1	5.9687	54.26	54.26
PC2	2.1639	19.67	73.93
PC3	1.0652	9.68	83.62
PC4	0.7388	6.72	90.33
PC5	0.4002	3.64	93.97
PC6	0.3356	3.05	97.02

Table 5: Environmental PCA variable scores, sorted by absolute PC1 loading.

Code	Variable	PC1	PC2
nit	Nitrate	1.150	-0.100
dfs	Distance from source	1.087	0.534
pho	Phosphate	1.058	-0.601
alt	Altitude	-1.044	-0.615
amm	Ammonium	1.007	-0.697
oxy	Dissolved oxygen	-0.975	0.499
bod	Biological oxygen demand	0.973	-0.715
flo	Flow rate	0.958	0.661
har	Water hardness	0.901	0.585
slo	Channel slope	-0.577	-0.489
pH	pH	-0.064	0.463

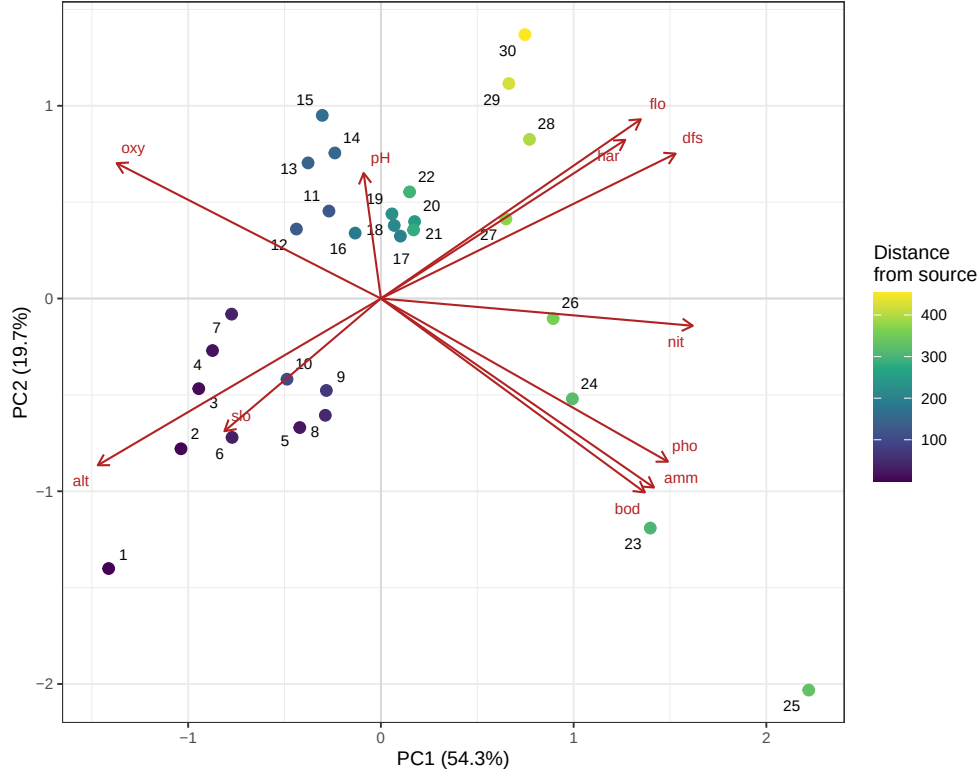


Figure 2: PCA biplot of the Doubs environmental variables. Arrows indicate environmental loadings; sites are coloured by distance from source.

3.2 Species CA and DCA

CA gives the same gradient through the fish assemblage. CA1 explains 51.5% of total inertia and CA2 explains 12.4%. DCA estimates the first gradient length as 3.86 standard deviation units, which is long enough for clear species turnover and explains why a CA arch appears on the second axis.

Table 6: Correspondence analysis eigenvalues.

Axis	Eigenvalue	Percent	Cumulative
CA1	0.6010	51.50	51.50
CA2	0.1444	12.37	63.87
CA3	0.1073	9.19	73.07
CA4	0.0834	7.14	80.21
CA5	0.0516	4.42	84.63
CA6	0.0418	3.59	88.22

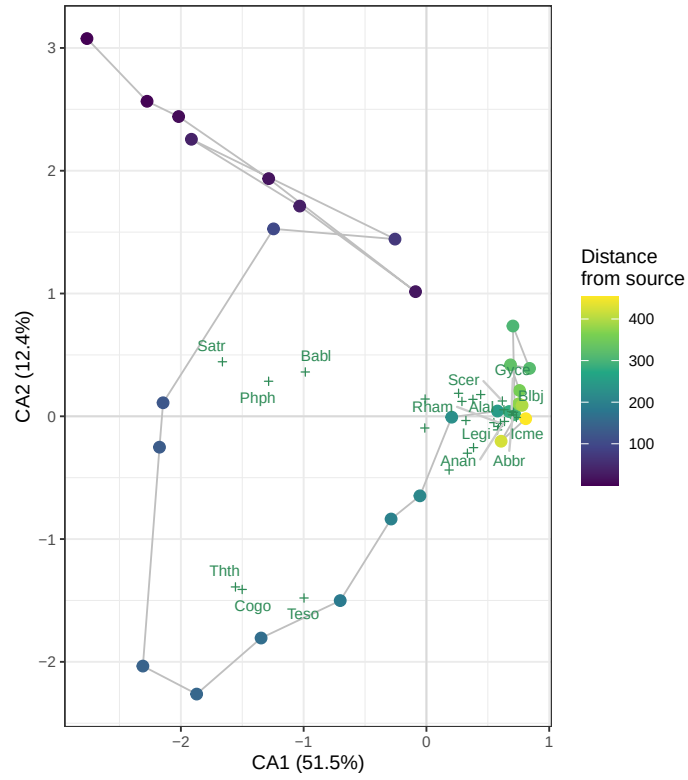


Figure 3: CA of Doubs fish abundances. Species labels are shown for the more peripheral taxa; points are sites coloured by distance from source.

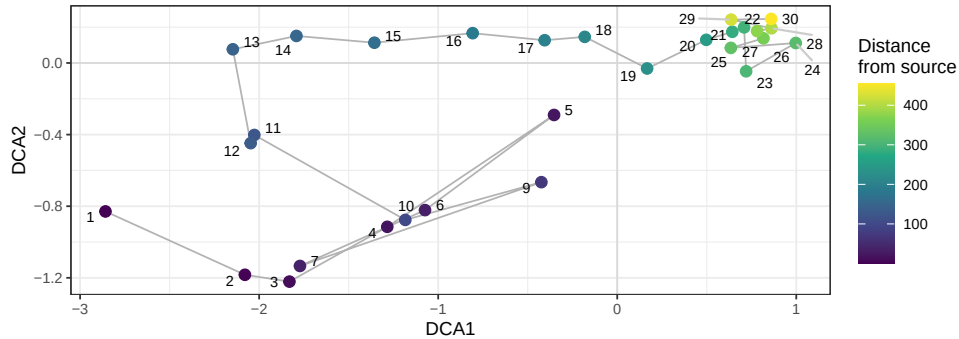


Figure 4: DCA of the Doubs fish data. The connected site sequence shows the long compositional gradient from source to lower reaches.

3.3 PCoA and nMDS

The distance-based ordinations recover the same gradient under a geometry free of the CA arch. Bray-Curtis PCoA MDS1 explains 52.4% of positive inertia, and the nMDS stress is 0.075, a good two-dimensional representation of the rank-order dissimilarities. Table 8 links the ordination figures

Table 7: PCoA positive eigenvalues for Bray-Curtis fish dissimilarities. Percentages are relative to the sum of positive eigenvalues.

Axis	Eigenvalue	Percent	Cumulative
MDS1	3.6953	52.35	52.35
MDS2	1.0985	15.56	67.92
MDS3	0.7105	10.07	77.98
MDS4	0.4150	5.88	83.86
MDS5	0.3046	4.31	88.18
MDS6	0.1918	2.72	90.89

to the ecological interpretation. For each unconstrained ordination it takes the site scores on the first axis and correlates them with distance from source, the one variable that orders the sites unambiguously from headwater to lower reach. The correlations are alignment diagnostics rather than model tests, since distance from source is fitted in none of these ordinations. It is projected onto them afterwards. A high value therefore says that an axis extracted from the data alone, whether from environmental covariance (PCA), fish turnover (CA), or fish dissimilarity (PCoA and nMDS), reconstructs the physical river order without being told to do so. The four correlations lie between 0.81 and 0.85, so the same upstream-to-downstream sequence emerges from every geometry and

from both the environmental and the faunal table. The dominant structure is a property of the system, not of any one method.

Table 8: Alignment between first ordination axes and the longitudinal river gradient.

Ordination	Correlation with distance from source
PCA PC1	0.853
CA CA1	0.812
PCoA MDS1	0.847
nMDS NMDS1	0.853

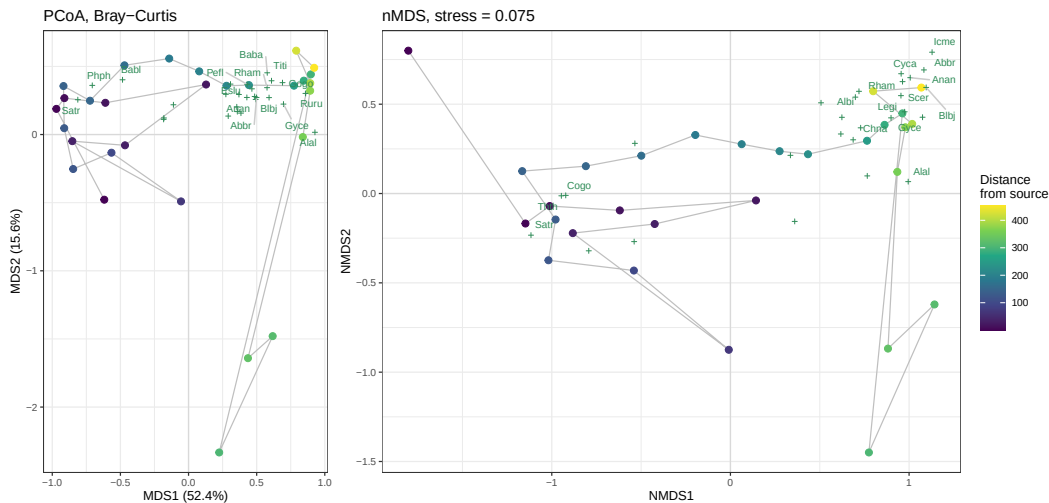


Figure 5: Bray-Curtis PCoA and nMDS ordinations of fish composition. Both recover the upstream-to-downstream species gradient.

3.4 Environmental Vector Fitting

Vector fitting projects each environmental variable a posteriori onto the CA of fish composition and identifies the variables most aligned with the assemblage pattern. The step is indirect-gradient in character. The CA is built from the fish matrix alone, and the environmental variables enter only afterwards, as arrows fitted to the finished configuration. Table 9 is therefore an interpretation aid and carries none of the inferential weight of a constrained model. Three quantities describe each fitted vector. Its direction gives the part of the ordination in which the variable increases, its r^2 gives the strength of association with the two-axis configuration, and its permutation p-value asks whether that association exceeds what random reassignment of the variable to sites would produce. Altitude ($r^2 = 0.81$), distance from source (0.69), and oxygen (0.63) give the longest vectors, followed by hardness, nitrate, and flow. These vectors share a common direction because they express one river-continuum gradient rather than several independent mechanisms. Ammonium, phosphate, and pH fit weakly ($r^2 \leq 0.18$), and two of them fail the permutation test, the first sign that the enrichment variables carry structure the continuum axis does not fully absorb.

Table 9: Environmental vectors fitted to the CA ordination. The r^2 column is the squared correlation between the fitted vector and the ordination configuration.

Code	Variable	Axis1	Axis2	r2	p
alt	Altitude	0.729	0.525	0.808	0.001
dfs	Distance from source	-0.787	-0.264	0.689	0.001
oxy	Dissolved oxygen	0.739	-0.284	0.626	0.001
har	Water hardness	-0.493	-0.479	0.472	0.001
nit	Nitrate	-0.637	-0.212	0.451	0.001
flo	Flow rate	-0.619	-0.248	0.444	0.001
slo	Channel slope	0.402	0.368	0.298	0.002
bod	Biological oxygen demand	-0.445	0.160	0.224	0.036
pho	Phosphate	-0.419	0.019	0.176	0.073
amm	Ammonium	-0.409	0.093	0.176	0.075
pH	pH	0.153	-0.260	0.091	0.231

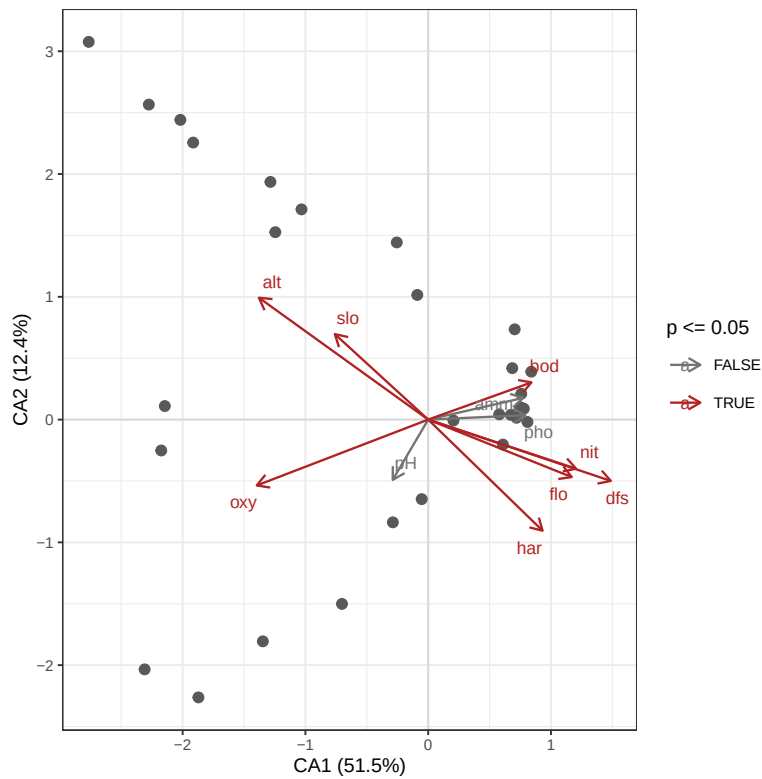


Figure 6: CA ordination with fitted environmental vectors. Red vectors are significant at $p \leq 0.05$.

4 Constrained Ordination

The constrained ordinations ask whether the environmental table explains the fish composition. They are direct-gradient analyses, since the environmental variables build the ordination axes and permutation tests then ask whether the explained component exceeds what chance would produce. This is a stronger statement than vector fitting, though still short of a proof of causation. In a river, many predictors are collinear because altitude, flow, distance from source, oxygen, nutrients, and organic load all change together along the channel. A constrained model can therefore show that the environmental matrix explains community structure while remaining unable to separate

the causal contribution of each correlated variable. Variance inflation factors (VIFs) and forward selection are two diagnostics applied before any interpretation of the selected models. A high VIF marks a variable that is largely predictable from the others, so it cannot be interpreted as an independent effect. Forward selection with `ordiR2step()` answers a different question. It starts from a null model, tries the candidate variables from the full model, adds the one that gives the strongest permitted gain in adjusted R^2 , and stops when no remaining candidate passes both the permutation test and the adjusted- R^2 ceiling. VIF is a diagnostic of predictor structure and `ordiR2step()` is response-informed selection, so the two are complementary. The selected variables describe the river-continuum gradient compactly rather than naming independent causal drivers. Table 10 summarises the three selected constrained models. Compare them on adjusted R^2 , which corrects for the number of predictors and so does not reward a model for merely carrying more variables. The raw R^2 would. The whole-model permutation p-values, all 0.001, show that each selected environmental set explains a detectable component of fish-community structure. The three models were built under different geometries and on different response transformations, yet they converge. RDA and db-RDA both retain `dfs`, `oxy`, and `bod`, differing only in the order of the last two, and CCA retains `dfs`, `oxy`, and `alt`. The convergence carries more weight than the small differences between models, because every model keeps one longitudinal-position variable (distance from source or altitude) together with oxygen and a marker of organic loading. Adjusted R^2 ranges from 0.50 in CCA to 0.57 in db-RDA, so roughly half of the community variation is captured by three correlated descriptors of river position. Table 11 decomposes each selected model term by

Table 10: Selected constrained ordination models.

Model	Selected predictors	Adjusted R^2	Overall F	Overall p
RDA	<code>dfs</code> , <code>oxy</code> , <code>bod</code>	0.536	11.79	0.001
CCA	<code>dfs</code> , <code>oxy</code> , <code>alt</code>	0.497	10.13	0.001
db-RDA	<code>dfs</code> , <code>bod</code> , <code>oxy</code>	0.570	13.36	0.001

term. The tests are sequential, applied in the order `ordiR2step()` added the variables, so each term is assessed over and above those already in the model. The first term absorbs the broadest gradient and later terms explain only what remains. Distance from source therefore carries by far the largest F statistic in every model (23.5 in RDA, 19.6 in CCA, 25.1 in db-RDA), since it stands for the dominant longitudinal ordering of the river. Oxygen and biological oxygen demand enter next and stay individually significant ($p \leq 0.003$), which means they capture chemical and organic-loading structure that distance from source alone leaves unexplained. That residual signal is the quantitative trace of the mid-river oxygen sag examined later in the report. Table 12 tests

Table 11: Permutation tests for terms in selected constrained models.

Model	Term	Df	Metric	Value	F	p
RDA	<code>dfs</code>	1	Variance	0.1959	23.54	0.001
RDA	<code>oxy</code>	1	Variance	0.0594	7.14	0.003
RDA	<code>bod</code>	1	Variance	0.0392	4.71	0.003
CCA	<code>dfs</code>	1	Chi-square	0.4120	19.56	0.001
CCA	<code>oxy</code>	1	Chi-square	0.1383	6.57	0.002
CCA	<code>alt</code>	1	Chi-square	0.0900	4.27	0.012
db-RDA	<code>dfs</code>	1	Sum of squares	2.7221	25.10	0.001
db-RDA	<code>bod</code>	1	Sum of squares	0.9277	8.55	0.001
db-RDA	<code>oxy</code>	1	Sum of squares	0.6974	6.43	0.001

the constrained axes rather than the named predictors. A constrained axis is a composite gradient, a weighted combination of the selected predictors ranked by the explained community variation it

carries. A significant first axis, with F between 22.9 and 28.5 across the three models, shows that the primary fitted environmental gradient is strong. Significant second and third axes show that the selected predictors retain lower-dimensional structure once that primary gradient is removed. These later axes reward caution, since each mixes several correlated variables and need not correspond to any single measured gradient.

Table 12: Permutation tests for constrained axes.

Model	Term	Df	Metric	Value	F	p
RDA	RDA1	1	Variance	0.2186	26.27	0.001
RDA	RDA2	1	Variance	0.0501	6.26	0.001
RDA	RDA3	1	Variance	0.0258	3.34	0.001
CCA	CCA1	1	Chi-square	0.4822	22.89	0.001
CCA	CCA2	1	Chi-square	0.1005	4.75	0.002
CCA	CCA3	1	Chi-square	0.0577	3.30	0.002
db-RDA	CAP1	1	Sum of squares	3.0894	28.49	0.001
db-RDA	CAP2	1	Sum of squares	0.9381	9.00	0.001
db-RDA	CAP3	1	Sum of squares	0.3197	3.18	0.004

4.1 VIF and Forward Selection Diagnostics

The full models deliberately begin with all measured environmental variables. This is useful because it shows the scale of the collinearity problem before any selection is done. The full-model VIFs are high for distance from source, altitude, flow, phosphate, ammonium, nitrate, oxygen, and biological oxygen demand, confirming that the river variables are not independent measurements. They are linked parts of the same physical and chemical continuum. This is why the report does not interpret the full-model coefficients or arrows as separate effects. After forward selection, the VIFs fall substantially. This does not make the selected variables causally independent, but it does mean the reduced models are less redundant and easier to interpret. The remaining VIFs are acceptable for a compact explanatory summary of the fish-community gradient. The forward-selection table records the actual `ordiR2step()` path. Each row is one accepted addition to the model. The adjusted R^2 column shows the cumulative explanatory power after that step, so it should increase from top to bottom within each model. The p-value tests the additional term at the point it enters the model, not in isolation from all other variables.

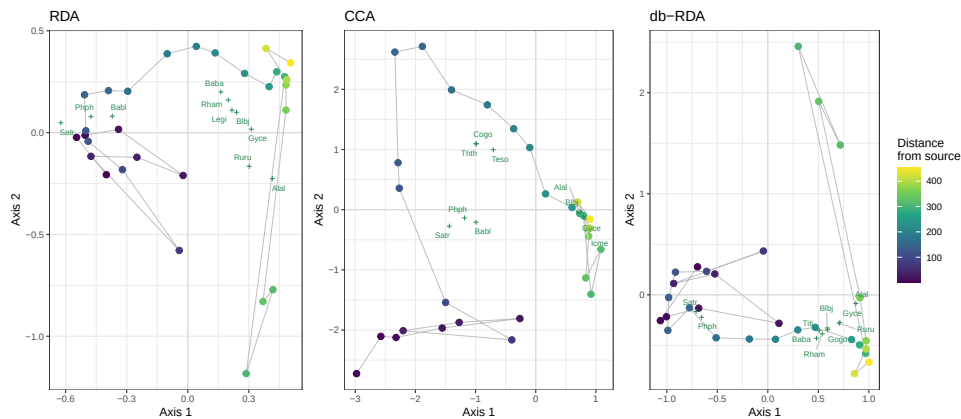


Figure 7: Selected RDA, CCA, and db-RDA models. Sites are coloured by distance from source.

Table 13: Variance inflation factors for the full constrained models before forward selection.

Model	Variable	Description	VIF
CCA	dfs	Distance from source	135.33
CCA	flo	Flow rate	54.01
CCA	alt	Altitude	31.38
CCA	oxy	Dissolved oxygen	13.48
CCA	amm	Ammonium	11.55
CCA	nit	Nitrate	10.53
CCA	bod	Biological oxygen demand	9.90
CCA	pho	Phosphate	9.47
CCA	har	Water hardness	4.25
CCA	slo	Channel slope	1.52
CCA	pH	pH	1.46
db-RDA	dfs	Distance from source	103.61
db-RDA	alt	Altitude	37.93
db-RDA	flo	Flow rate	34.97
db-RDA	amm	Ammonium	30.98
db-RDA	pho	Phosphate	27.21
db-RDA	bod	Biological oxygen demand	18.02
db-RDA	nit	Nitrate	16.95
db-RDA	oxy	Dissolved oxygen	12.99
db-RDA	har	Water hardness	3.99
db-RDA	slo	Channel slope	2.05
db-RDA	pH	pH	1.32
RDA	dfs	Distance from source	103.61
RDA	alt	Altitude	37.93
RDA	flo	Flow rate	34.97
RDA	amm	Ammonium	30.98
RDA	pho	Phosphate	27.21
RDA	bod	Biological oxygen demand	18.02
RDA	nit	Nitrate	16.95
RDA	oxy	Dissolved oxygen	12.99
RDA	har	Water hardness	3.99
RDA	slo	Channel slope	2.05
RDA	pH	pH	1.32

Table 14: Variance inflation factors for the selected constrained models after forward selection.

Model	Variable	Description	VIF
CCA	dfs	Distance from source	8.97
CCA	alt	Altitude	7.19
CCA	oxy	Dissolved oxygen	1.73
db-RDA	oxy	Dissolved oxygen	4.12
db-RDA	bod	Biological oxygen demand	3.43
db-RDA	dfs	Distance from source	1.50
RDA	oxy	Dissolved oxygen	4.12
RDA	bod	Biological oxygen demand	3.43
RDA	dfs	Distance from source	1.50

Table 15: Forward-selection paths from `ordiR2step()` for the constrained ordinations.

Model	Step	Added variable	Adjusted R2 after step	F	p
RDA	1	dfs	0.367	17.25	0.001
RDA	2	oxy	0.470	6.25	0.001
RDA	3	bod	0.536	4.71	0.003
CCA	1	dfs	0.330	14.74	0.001
CCA	2	oxy	0.433	5.83	0.006
CCA	3	alt	0.496	4.27	0.009
db-RDA	1	dfs	0.363	16.95	0.001
db-RDA	2	bod	0.480	7.08	0.001
db-RDA	3	oxy	0.570	6.43	0.002

4.2 Data-driven and theory-driven variable selection

Forward selection is a statistical procedure, not an ecological hypothesis. It keeps whichever variables most improve adjusted R^2 , and in these data that is `dfs` together with `oxy` and `bod`. A mechanistic reading of the same river suggests a different starting point. Position in the catchment sets the physical template, since altitude and distance from source govern temperature, slope, and discharge. Land use along the valley then loads the water with dissolved inorganic nitrogen (DIN, namely nitrate plus ammonium) and with phosphate. That nutrient load drives primary production and the supply of degradable organic matter, which raises biological oxygen demand, and the respiration that follows draws down dissolved oxygen. In this chain nutrients are the upstream lever and oxygen is a downstream consequence. A researcher who wants the model to name a cause rather than a correlate might therefore prefer a nutrient variable over oxygen, on the grounds that nutrient input is what a catchment manager can actually change. Table 16 puts that intuition to a test. It fits several candidate Hellinger RDA models and compares their adjusted R^2 and their worst variance inflation factor. Distance from source alone already reaches adjusted $R^2 = 0.37$. Adding nitrate lifts it only to 0.39, whereas adding oxygen and biological oxygen demand lifts it to 0.54. The two nutrient-loading models, altitude with nitrate and either ammonium or phosphate, settle near 0.39 as well, and their variance inflation factors stay modest. Swapping the enrichment pair for a nutrient variable therefore costs about 0.15 of adjusted R^2 , which is not a rounding difference. The reason for that cost is visible in the correlations of Table 17. Along the clean part

Table 16: Data-driven and theory-driven candidate constrained models, all fitted as Hellinger RDA. Adjusted R^2 measures explanatory power and the maximum VIF measures residual collinearity within each candidate set.

Model	Predictors	Rationale	Adjusted R2	Max VIF
Forward-selected	<code>dfs</code> , <code>oxy</code> , <code>bod</code>	<code>ordiR2step()</code> data-driven selection	0.536	4.12
Position only	<code>dfs</code>	Single longitudinal-position variable	0.367	1.00
Position + nitrate	<code>dfs</code> , <code>nit</code>	Position plus the nutrient that tracks the continuum	0.388	2.20
Nutrient loading	<code>alt</code> , <code>nit</code> , <code>amm</code>	Altitude plus dissolved inorganic nitrogen	0.389	8.14
Nutrient + phosphate	<code>alt</code> , <code>nit</code> , <code>pho</code>	Altitude plus nitrogen and phosphorus loading	0.383	6.32

of the continuum, oxygen, biological oxygen demand, nitrate, ammonium, and phosphate move together, with many pairwise correlations above 0.7 in absolute value, so any one of them stands in for the others and the choice among them is largely interpretive. The exception is where the ecology becomes interesting. Nitrate rises almost monotonically with distance from source ($r = 0.75$) and tracks the continuum, whereas oxygen and biological oxygen demand decouple from it at the mid-river reach near sites 23 to 25, where oxygen falls to 4 to 6 mg L⁻¹ and biological oxygen demand exceeds 12 mg L⁻¹. Forward selection retains oxygen and biological oxygen demand precisely because they carry that local oxygen sag, which a position or nitrate model cannot reproduce. The honest reading is therefore layered. For a management question about nutrient control, DIN is the defensible predictor, and it forfeits little explanatory power over the continuum. For describing the community that is actually present, oxygen is more than a proxy; near the pollution reach it is a proximate stressor with a signal of its own, and dropping it would discard the one place where the river departs from a smooth gradient. Statistical selection cannot arbitrate between these readings, because the variables are collinear over most of the river and diverge only at a handful of sites. The choice is an ecological judgement informed by the correlation structure, not an output of the algorithm. This is the sense in which explanation is not causation: `ordiR2step()` finds the most economical description of the pattern, but the causal ordering of nutrients, organic load, and oxygen

Table 17: Pearson correlations among distance from source and the enrichment variables across the 29 matched sites. The tight cluster among oxygen, biological oxygen demand, nitrate, ammonium, and phosphate is the collinearity that makes them substitutable along the continuum.

	dfs	oxy	bod	nit	amm	pho
dfs	1.00	-0.57	0.43	0.74	0.41	0.47
oxy	-0.57	1.00	-0.84	-0.69	-0.75	-0.76
bod	0.43	-0.84	1.00	0.68	0.90	0.91
nit	0.74	-0.69	0.68	1.00	0.80	0.80
amm	0.41	-0.75	0.90	0.80	1.00	0.97
pho	0.47	-0.76	0.91	0.80	0.97	1.00

must be supplied by the ecology.

5 Additional Advanced Analyses

The analyses in this section sit outside the core Quantitative Ecology chapter sequence. Each re-examines the river-continuum interpretation from an angle the ordinations do not cover, namely the separation of environment from space, the correlation of whole dissimilarity matrices, the decomposition of beta diversity, the identification of ecologically distinctive sites and species, and the shape of individual species responses. They extend the ordinations rather than replace them, and several of them sharpen the interpretation by showing where the smooth-continuum picture breaks down. Table 18 maps the section.

Table 18: Advanced analyses added beyond the core Quantitative Ecology workflow, with their purpose and headline result.

Analysis	Purpose	Result
Variation partitioning	Separate pure environmental, pure spatial, and shared spatial-environmental fractions.	Environment only adj. $R^2 = 0.106$; space only adj. $R^2 = 0.123$
MEM spatial eigenvectors	Model spatially structured community variation using Moran eigenvector maps.	MEM2, MEM3, MEM1, MEM4; adj. $R^2 = 0.551$; $p = 0.001$
Procrustes comparison	Compare environmental PCA geometry with fish-community nMDS geometry.	Procrustes $r = 0.748$; $p = 0.001$
Negative-binomial manyGLM	Test species abundances with a model-based multivariate count framework.	dfs, bod, and oxy $p = 0.002, 0.002, 0.002$
Mantel and partial Mantel	Correlate the fish, environmental, and geographic distance matrices directly.	partial env space $r = 0.564$; partial space env $r = 0.177$
Beta-diversity partitioning	Split total dissimilarity into species replacement and abundance-gradient parts.	turnover = 0.86 of total; gradient = 0.14
LCBD and SCBD	Rank sites and species by their contribution to total beta diversity.	5 sites at $p \leq 0.05$, incl. headwater and pollution reach
Species-response GLMs	Fit abundance responses to distance from source for individual species.	5 of 8 species unimodal within survey

5.1 Environment-Space Variation Partitioning

Variation partitioning splits the Hellinger-transformed community variation into four fractions, namely the part explained uniquely by the selected environmental variables, the part explained uniquely by spatial position, the part they explain jointly, and the unexplained remainder. The exercise is informative here because river environments are spatially organised. Sites further downstream are also further along the channel, so environment and space are confounded by construction, and a large shared fraction is expected rather than a nuisance to be removed. Working through Table 19, the pure environmental fraction is adjusted $R^2 = 0.11$ and the pure spatial fraction is 0.12, while the shared fraction reaches 0.43. Most of the explained community variation is therefore jointly attributable to environment and position, which is the numerical statement of the fact that the environmental gradient is laid out along the length of the river. Only about a third of the variation (adjusted $R^2 = 0.34$) is left unexplained.

Table 19: Variation partitioning between selected environmental predictors and polynomial spatial coordinates.

Fraction	Interpretation	Adjusted R2	p
Environment only	Selected environmental predictors after conditioning on polynomial space	0.106	0.001
Space only	Spatial trend after conditioning on selected environmental predictors	0.123	0.001
Shared environment-space	Variation jointly structured by measured environment and spatial position	0.430	
Residual	Unexplained variation	0.341	

5.2 Spatial Eigenvectors, Procrustes Rotation, and manyGLM

Moran eigenvector maps (MEMs) are orthogonal spatial variables that describe pattern at a range of scales, from a broad upstream-to-downstream trend to finer alternations between neighbouring reaches. Forward selection over the MEM set retains four of them and reaches adjusted $R^2 = 0.55$ ($p = 0.001$), which confirms that the fish assemblage is spatially structured along the river network. The MEMs name no environmental cause. They quantify the scale at which spatial structure exists, and the broadest-scale MEMs carry the same longitudinal signal as distance from source. The Procrustes rotation asks a geometric question. It rotates and rescales the environmental PCA configuration onto the fish-community nMDS configuration and measures how closely the two point clouds coincide. The fit is strong (Procrustes $r = 0.75$, $p = 0.001$), so a site that is unusual in environmental space tends to be unusual in the same way in faunal space. The two ordinations describe one gradient in two currencies. The manyGLM analysis moves from distance-based to model-based inference. It fits a negative-binomial generalised linear model to each species, respecting the mean-variance relationship of counts that dissimilarity indices only approximate, and it tests the selected predictors jointly by resampling (Table 20). Distance from source, biological oxygen demand, and oxygen are each significant ($p = 0.002$), with distance from source carrying much the largest deviance, 430 against 133 and 112 for the two chemistry terms. A method that makes the count distribution explicit therefore reaches the same ranking as the ordinations, which is reassurance that the pattern does not depend on the Bray-Curtis geometry. The agreement across MEM, Procrustes, and manyGLM supports the main conclusion and repeats the standing caution that environment and space are tightly coupled here.

Table 20: Negative-binomial manyGLM multivariate tests for the db-RDA selected predictors.

Term	Residual df	Df diff	Deviance	p
dfs	27	1	430.2	0.002
bod	26	1	133.2	0.002
oxy	25	1	112.2	0.002

5.3 Matrix Correlation with Mantel and Partial Mantel Tests

The Mantel test correlates two distance matrices directly, without first reducing either to ordination axes. Here it compares the fish Bray-Curtis dissimilarity with a Euclidean distance on the standardised environmental variables, and with the geographic distance between sites (Table 21). Both simple correlations are significant, but the environmental matrix is much the stronger match (Mantel $r = 0.61$ against $r = 0.32$ for geography). The partial tests then hold one matrix constant while testing the other. Community structure remains strongly tied to environment once geography is held constant ($r = 0.56$, $p = 0.001$), whereas its tie to geography almost disappears once environment is held constant ($r = 0.18$, $p = 0.015$). The environment carries most of the signal,

and the pure-space contribution is small. This agrees in direction with the variation partitioning, though the two methods weight the environment differently, since the Mantel test uses the full eleven-variable environmental distance while the partitioning uses only the three selected predictors. Mantel correlations are also known to be conservative and to depend on the distance measure, so they are read as corroboration rather than as a separate estimate of effect size.

Table 21: Mantel and partial Mantel correlations among the fish, environmental, and geographic distance matrices, with 999 permutations.

Test	Mantel r	p
Community vs environment	0.606	0.001
Community vs geographic distance	0.319	0.001
Community vs environment, geography held constant	0.564	0.001
Community vs geography, environment held constant	0.177	0.015

5.4 Beta-Diversity Components: Replacement and Abundance Gradient

The central claim of the report is that downstream change is a replacement of one set of species by another rather than a simple loss of species. Beta-diversity partitioning tests that claim quantitatively by splitting the total multi-site Bray-Curtis dissimilarity into a balanced-variation component, which is true turnover where a gain of one species is matched by a loss of another, and an abundance-gradient component, which is the nestedness-like part where assemblages differ because one is a subset or a diluted version of the other (Table 22). Turnover dominates. It accounts for 0.86 of the total dissimilarity, against 0.14 for the abundance-gradient component. The pattern is therefore replacement, which is consistent with the long DCA gradient and with the species arch in the CA. The abundance-gradient component is not zero, and the sites that inflate it are the depauperate ones at the two ends of the disturbance range, namely the headwater and the polluted mid-river reach, where the assemblage is a thinned subset of the richer lowland fauna rather than a distinct set of species. Beta-diversity partitioning thus recovers both processes and shows which one dominates.

Table 22: Multi-site beta-diversity partitioning of the fish assemblage into balanced turnover and abundance-gradient components (Baselga family, abundance-based).

Component	Value	Share of total
Total dissimilarity (beta.BRAY)	0.909	1.000
Balanced turnover (species replacement)	0.779	0.857
Abundance gradient (nestedness-like)	0.130	0.143

5.5 Distinctive Sites and Species: LCBD and SCBD

Local contributions to beta diversity (LCBD) rank the sites by how much each one adds to the total variance of the community matrix, and species contributions (SCBD) do the same for the species. High-LCBD sites are ecologically distinctive, whether through rarity, unusual composition, or a departure from the general trend. Table 23 lists the six sites with the largest LCBD, and the permutation p-values flag those whose distinctiveness exceeds chance. The result is telling. The most distinctive sites are the extreme headwater (site 1, a near-monospecific brown-trout reach) and the organic-pollution zone (sites 23 to 25), each significant at $p \leq 0.05$. These are exactly the

two places where the smooth continuum is interrupted, the first by the severity of the upstream physical filter and the second by the mid-river oxygen sag. The heaviest SCBD contributions come from the gradient-defining species, led by **Satr** (Brown trout, 0.142), **Phph** (Minnow, 0.095), **Ala1** (Bleak, 0.080), **Bab1** (Stone loach, 0.071), **Ruru** (Roach, 0.055). Figure 8 places the LCBD values and the site-level species richness along distance from source, and the richness collapse at sites 23 to 25, from 22 species down to 3, coincides with the LCBD peak there.

Table 23: The six sites with the largest local contribution to beta diversity (LCBD), with distance from source and the permutation p-value.

Site	Distance from source	LCBD	p
1	0.3	0.0725	0.001
23	304.3	0.0616	0.001
24	314.7	0.0524	0.011
13	143.6	0.0477	0.044
25	327.8	0.0476	0.037
2	2.2	0.0445	0.094

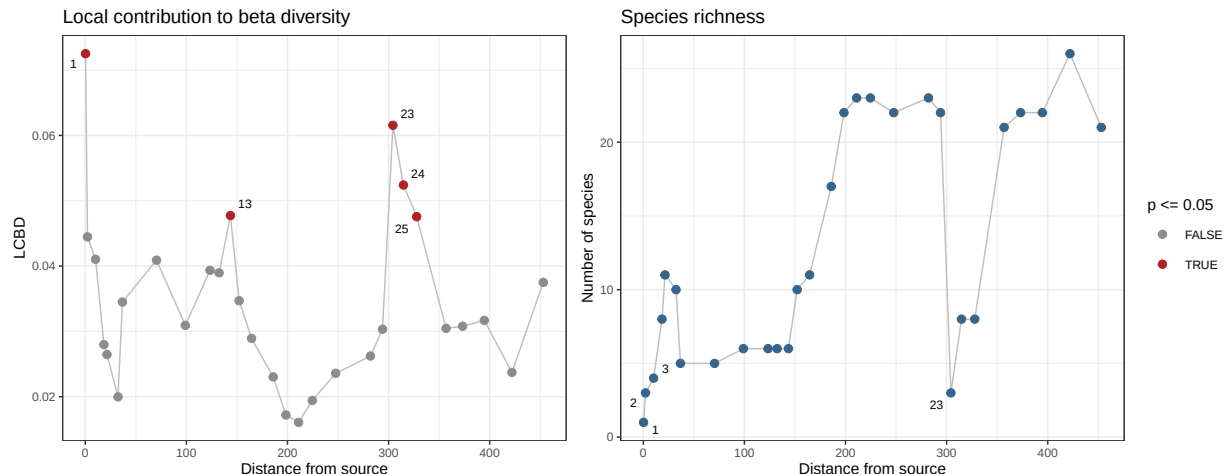


Figure 8: Local contribution to beta diversity (left) and species richness (right) along distance from source. Firebrick points on the left mark sites significant at $p \leq 0.05$; the richness minimum at sites 23 to 25 marks the organic-pollution reach.

5.6 Species-Response Shapes Along the Gradient

The ordinations assume, through the chi-square and Bray-Curtis geometries, that species rise and fall along the gradient rather than increasing without limit. A direct way to check that assumption is to fit each species against distance from source with a negative-binomial generalised linear model that includes a linear and a quadratic term, `glm.nb(y ~ dfs + I(dfs^2))`. A significant negative quadratic term indicates a unimodal response with an optimum inside the surveyed reach, and the optimum itself is the fitted peak position (Table 24, Figure 9). Headwater and mid-river species show clear unimodal responses with optima ordered along the channel, from brown trout near 50 km through stone loach, bullhead, and grayling to barbel near 350 km. The lowland species behave differently. Roach, gudgeon, and common bream are still rising at the downstream end of the survey, so their fitted optima lie beyond the sampled reach and their responses are monotonic

within it. This is the expected signature of a gradient sampled from its cold, species-poor source to a lower reach that has not yet reached the true lowland optima of the most downstream taxa, and it is the same truncation that gives the DCA its long first axis.

Table 24: Negative-binomial species-response models against distance from source. The fitted optimum is the peak of the quadratic response where it falls inside the surveyed reach; the quadratic p-value tests for curvature.

Code	Species	Fitted optimum (km)	Quadratic p	Response
Satr	Brown trout	53	0.022	Unimodal, optimum within survey
Cogo	Bullhead	163	0.020	Unimodal, optimum within survey
Thth	Grayling	182	0.016	Unimodal, optimum within survey
Babl	Stone loach	99	0.010	Unimodal, optimum within survey
Baba	Barbel	348	0.017	Unimodal, optimum within survey
Gogo	Gudgeon	beyond survey	0.317	Increasing, optimum below survey
Ruru	Roach	beyond survey	0.979	Increasing, optimum below survey
Abbr	Common bream	beyond survey	0.200	Increasing, optimum below survey

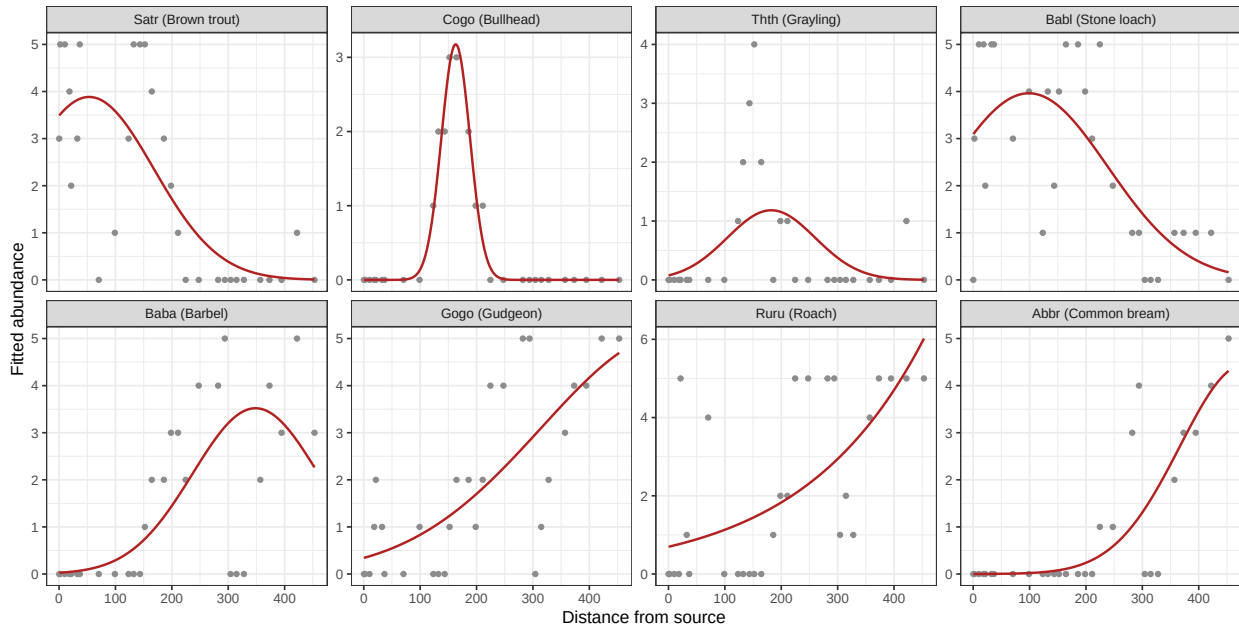


Figure 9: Fitted negative-binomial abundance responses (firebrick) with observed counts (grey) against distance from source, for species spanning the gradient. Vertical scales differ between panels.

6 Species Replacement Along the Gradient

The ordinations reduce the data to axes. The raw abundance pattern shows the ecological turnover directly (Figure 10). The gradient has a physical origin. Near the source the channel is high, steep, and cold, the discharge is low but turbulent, and the water is close to oxygen saturation. Downstream the profile flattens, discharge and hardness rise, the water warms, and organic matter accumulates and decays over a longer residence time. This physical template sets up a chemical one, since warmer, slower, nutrient-rich water supports more production and a higher biological

oxygen demand. The fish assemblage is layered onto both. Each species occupies a bounded stretch of the gradient, and the stretches are ordered, which is what produces replacement rather than simple enrichment or loss. The headwaters hold cool-water specialists, namely brown trout, bullhead, minnow, and grayling, whose fitted optima fall in the upper reaches (Table 24). Stone loach and barbel take over through the middle river. The lower reaches are held by cyprinids and generalists such as roach, common bream, bleak, and eel, whose abundances are still climbing at the downstream limit of the survey. The sequence recovers the classical river fish zonation, from a trout zone through grayling and barbel zones to a bream zone, expressed as a continuous replacement rather than a set of sharp boundaries. One reach departs from the smooth sequence. At sites 23 to 25 the water chemistry changes abruptly, with biological oxygen demand rising above 12 mg L^{-1} , ammonium reaching 1.8 mg L^{-1} , and dissolved oxygen falling to 4 to 6 mg L^{-1} . This is the chemical signature of a point-source organic load. Species richness collapses from 22 at site 22 to 3 at site 23 before recovering downstream (Figure 8), and these sites carry the highest local contributions to beta diversity outside the headwater. The organic-pollution reach is a second, shorter gradient superimposed on the longitudinal one, an oxygen sag that thins the assemblage locally and then releases it as the river re-aerates. This reach is what keeps oxygen and biological oxygen demand in the selected constrained models and what lifts the abundance-gradient component of beta diversity above zero.

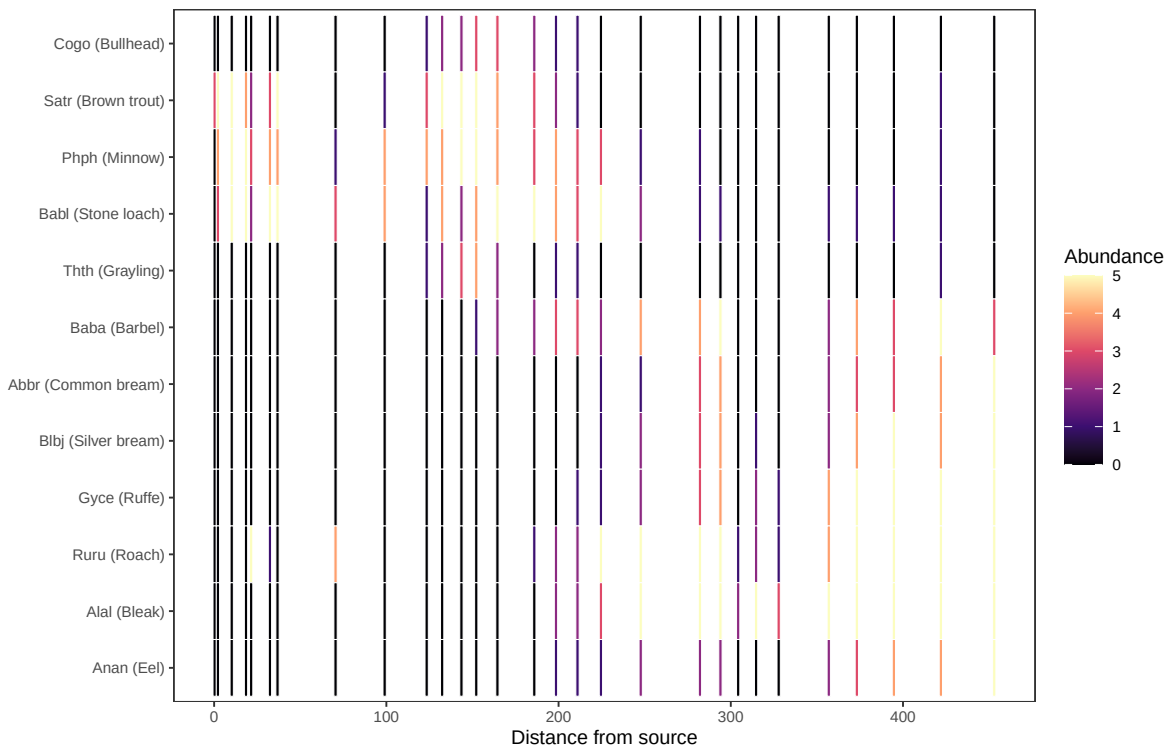


Figure 10: Abundance of representative fish species along distance from source. Species are ordered to show the broad upstream-to-downstream replacement.

7 Discussion: Gradients that Structure the System

The Doubs assemblage is organised first and foremost by a longitudinal river-continuum gradient. The upper sites are high, steep, cool reaches with near-saturated oxygen and little nutrient or

organic load. The lower sites are further from source and lower in altitude, with greater discharge, harder water, more nutrients, and higher biological oxygen demand. These variables correlate because they are different measurements of one downstream integration of catchment, geomorphological, and biogeochemical processes, and every analysis in this report, unconstrained or constrained, distance-based or model-based, recovers that single axis as the dominant structure. The fish assemblage is layered onto this template as an ordered replacement. The upstream end supports trout-zone species, and downstream those decline and give way to cyprinids and generalists of the warmer, slower, more productive lower river. Because each species occupies a bounded stretch of the gradient (Table 24), the turnover is genuine replacement rather than progressive enrichment, which the beta-diversity partitioning confirms by attributing 0.86 of total dissimilarity to balanced turnover. CA, PCoA, and nMDS recover the same sequence from composition alone, so the interpretation rests on the data rather than on one ordination geometry. The continuum is not perfectly smooth. A short reach at sites 23 to 25 carries a strong organic-load signal, with biological oxygen demand above 12 mg L^{-1} and oxygen down to 4 mg L^{-1} , and the fauna there thins to a few tolerant species before recovering downstream. This reach is the most distinctive part of the river after the headwater on the LCBD ranking, it supplies most of the nestedness-like component of beta diversity, and it is the reason oxygen and biological oxygen demand survive forward selection where nitrate, which tracks the continuum more faithfully, does not. Read together, the analyses describe two overlaid gradients, a long physical-chemical continuum and a short, sharp oxygen sag, rather than a single monotonic trend. The second ordination axes reward caution. CA2 in particular is partly the arch effect of a long first gradient rather than a separate ecological process, and DCA supports that reading with a first-axis length of 3.9 standard-deviation units. Where second-axis structure is genuine, it is best interpreted as local departure from the main sequence, and the oxygen-sag reach is the clearest example. The constrained models show that the measured environment explains about half of the community variation, yet they cannot rank the predictors causally, because those predictors are correlated expressions of river position and organic load. Forward selection returns the most economical description, namely distance from source with oxygen and biological oxygen demand, while a mechanistic account would place dissolved inorganic nitrogen upstream of the oxygen response. The two views are compatible. Statistics identifies the compact set of correlates, and ecology supplies the causal ordering. Neither on its own licenses a claim that a single variable drives the fish community. The analyses are correlational, so association is not mechanism. The gradient is truncated at both ends, so the optima of the most downstream species lie beyond the survey and the headwater is sampled to near-zero richness. Environment and space are confounded along a single channel, so the pure environmental and pure spatial fractions are both small while the shared fraction is large. These are properties of a one-dimensional river surveyed once, and they would be relaxed only by sampling more rivers, more seasons, or by manipulative work.

8 Reproducibility

This report was generated by `BCB743/reports/make_doubs_multivariate_report.R`. The run used `vegan` 2.7.5 under R 4.5.3, with 999 permutations for the ordination and matrix tests. The advanced analyses used `adespatial` 0.3.29, `betapart` 1.6.1, `mvabund` 4.2.8, `MASS` 7.3.65. All stochastic steps were seeded with `set.seed(743)`.